Packet Tracer - Modify Single-Area OSPFv2 (Instructor Version)

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only.

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
	S0/1/0	209.165.200.225	255.255.255.224	
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.10.6	255.255.255.252	
	S0/0/1	192.168.10.10	255.255.255.252	
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objectives

Part 1: Modify OSPF Default Settings

Part 2: Verify Connectivity

Scenario

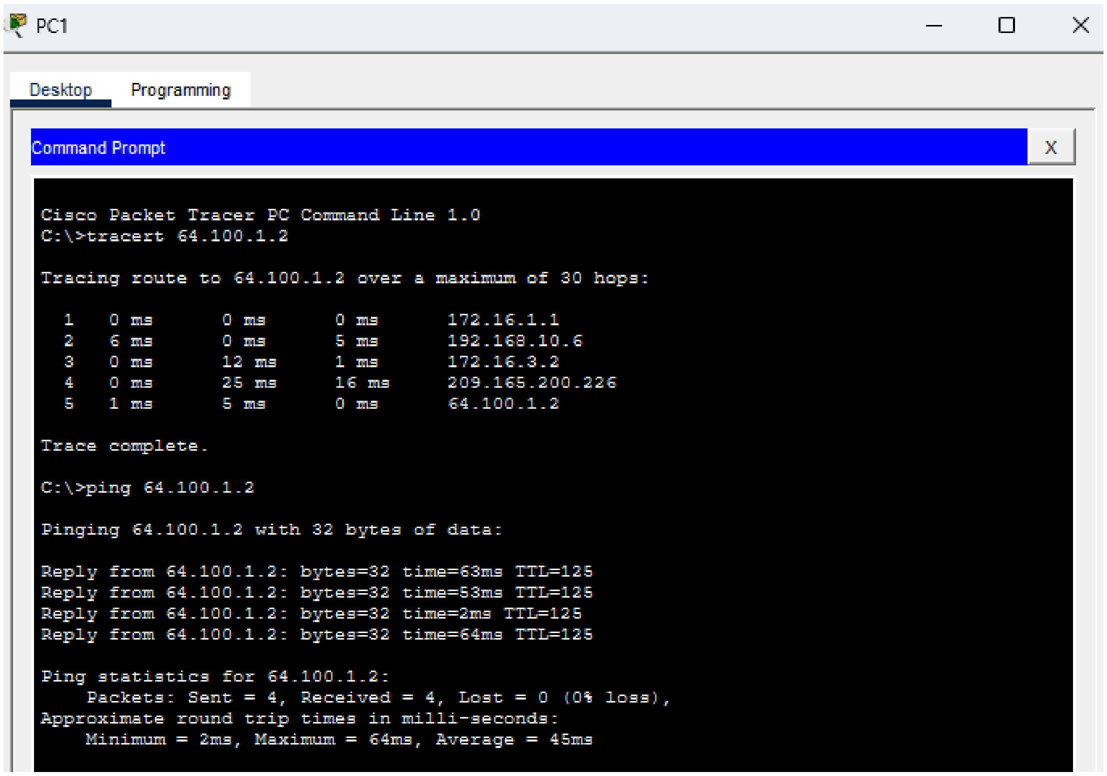
In this activity, OSPF is already configured and all end devices currently have full connectivity. You will modify the default OSPF routing configurations by changing the hello and dead timers and adjusting the bandwidth of a link. Then you will verify that full connectivity is restored for all end devices.

Instructions

Part 1: Modify OSPF Default Settings

Step 1: Test connectivity between all end devices.

Before modifying the OSPF settings, verify that all PCs can ping the web server and each other.




```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 64.100.1.2

Tracing route to 64.100.1.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.1.1
  2   6 ms      0 ms      5 ms      192.168.10.6
  3   0 ms      12 ms     1 ms      172.16.3.2
  4   0 ms      25 ms     16 ms     209.165.200.226
  5   1 ms      5 ms      0 ms      64.100.1.2

Trace complete.

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=63ms TTL=125
Reply from 64.100.1.2: bytes=32 time=53ms TTL=125
Reply from 64.100.1.2: bytes=32 time=2ms TTL=125
Reply from 64.100.1.2: bytes=32 time=64ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 64ms, Average = 45ms
```

Step 2: Adjust the hello and dead timers between R1 and R2.

a. Enter the following commands on **R1**.

```
R1(config)# interface s0/0/0
R1(config-if)# ip ospf hello-interval 15
R1(config-if)# ip ospf dead-interval 60
```

b. After a short period of time, the OSPF connection with **R2** will fail, as shown in the router output.

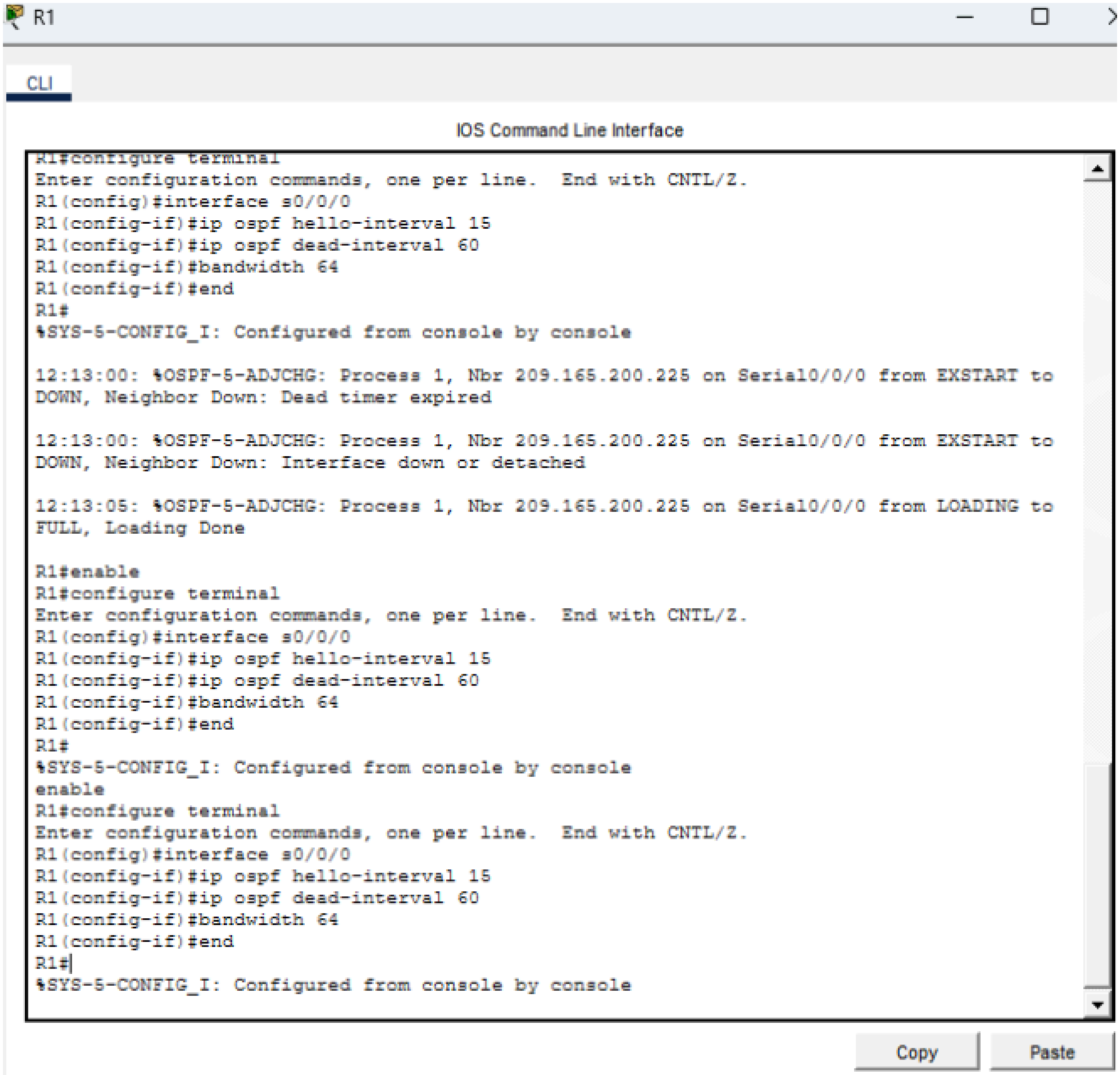
```
00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
```

Both sides of the connection need to have the same timer values in order for the adjacency to be maintained. Identify the interface on R2 that is connected to R1. Adjust the timers on the R2 interface to match the settings on **R1**.

```
R2(config)# interface s0/0/0
R2(config-if)# ip ospf hello-interval 15
R2(config-if)# ip ospf dead-interval 60
```

After a brief period of time you should see a status message that indicates that the OSPF adjacency has been reestablished.

```
00:21:52: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from
LOADING to FULL, Loading Done
```




```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s0/0/0
R1(config-if)#ip ospf hello-interval 15
R1(config-if)#ip ospf dead-interval 60
R1(config-if)#bandwidth 64
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

12:13:00: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from EXSTART to
DOWN, Neighbor Down: Dead timer expired

12:13:00: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from EXSTART to
DOWN, Neighbor Down: Interface down or detached

12:13:05: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from LOADING to
FULL, Loading Done

R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s0/0/0
R1(config-if)#ip ospf hello-interval 15
R1(config-if)#ip ospf dead-interval 60
R1(config-if)#bandwidth 64
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s0/0/0
R1(config-if)#ip ospf hello-interval 15
R1(config-if)#ip ospf dead-interval 60
R1(config-if)#bandwidth 64
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```


Step 3: Adjust the bandwidth setting on R1.

a. Trace the path between **PC1** and the web server located at 64.100.1.2. Notice that the path from **PC1** to 64.100.1.2 is routed through **R2**. OSPF prefers the lower cost path.

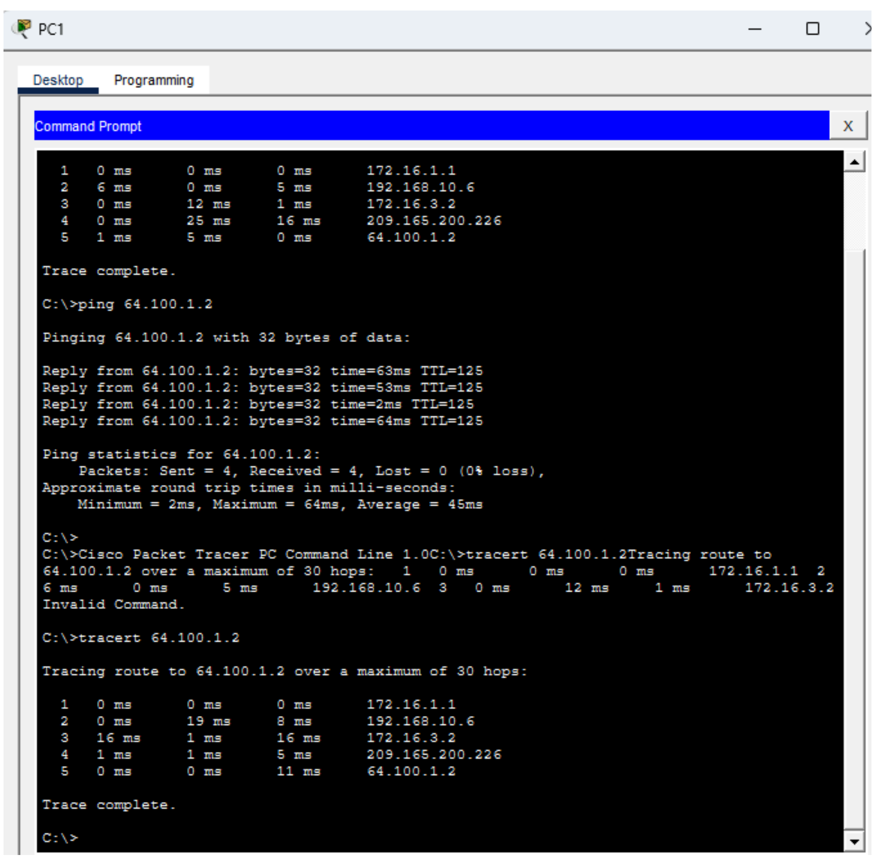
```
C:\> tracert 64.100.1.2
Tracing route to 64.100.1.2 over a maximum of 30 hops:
1 1 ms 0 ms 8 ms 172.16.1.1
2 0 ms 1 ms 0 ms 172.16.3.2
3 1 ms 9 ms 2 ms 209.165.200.226
4 * 1 ms 0 ms 64.100.1.2
Trace complete.
```

b. On the **R1** Serial 0/0/0 interface, set the bandwidth to 64 Kb/s. This does not change the actual port speed, only the metric that the OSPF process on **R1** will use to calculate best routes.

```
R1(config-if)# bandwidth 64
```

c. Trace the path between **PC1** and the web server located at 64.100.1.2. Notice that the path from **PC1** to 64.100.1.2 is redirected through **R3**. OSPF prefers the lower cost path.

```
C:\> tracert 64.100.1.2
Tracing route to 64.100.1.2 over a maximum of 30 hops:
1 1 ms 0 ms 3 ms 172.16.1.1
2 8 ms 1 ms 1 ms 192.168.10.6
3 2 ms 0 ms 2 ms 172.16.3.2
4 2 ms 3 ms 1 ms 209.165.200.226
5 2 ms 11 ms 11 ms 64.100.1.2
Trace complete.
```



Part 2: Verify Connectivity

Verify that all PCs can ping the web server and each other.

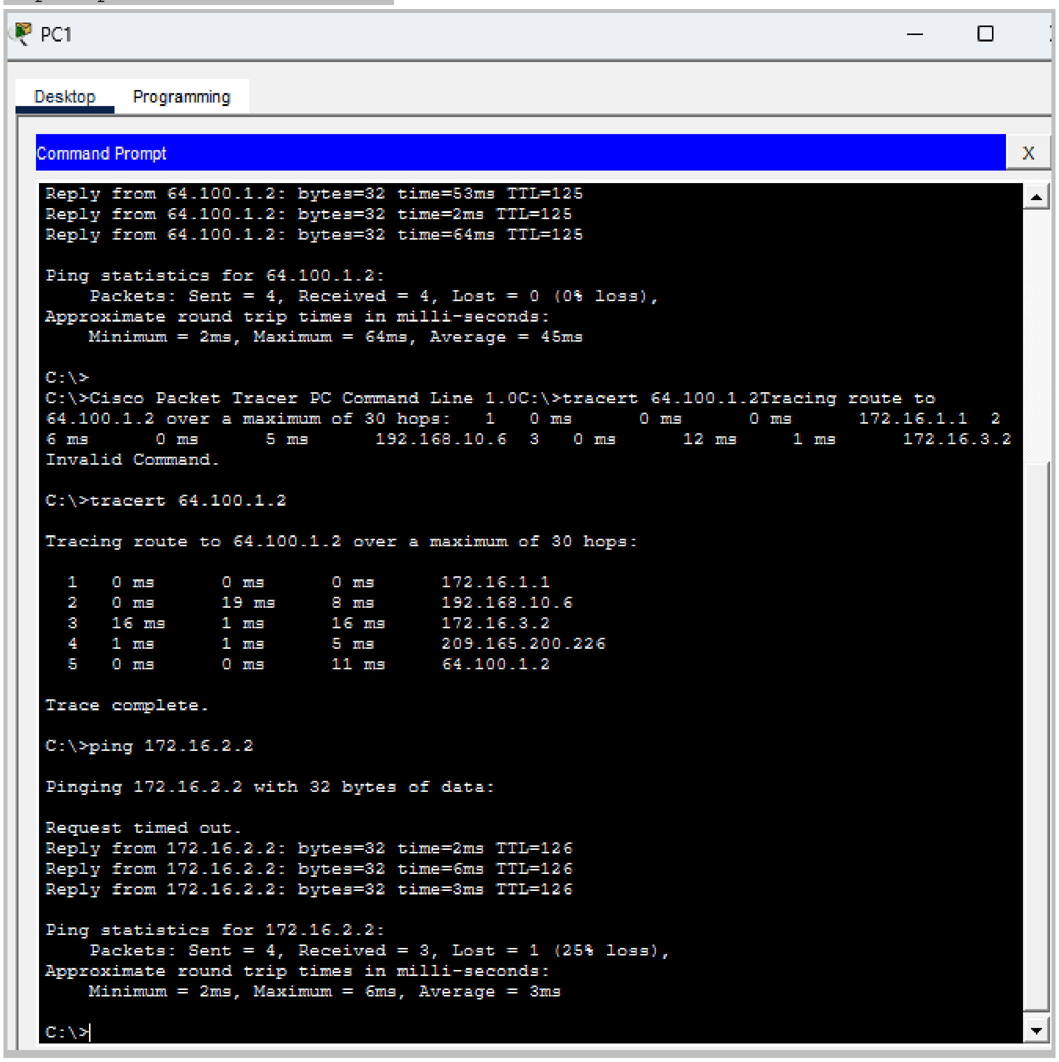
Answer scripts

Router R1

```
interface s0/0/0
 ip ospf hello-interval 15
 ip ospf dead-interval 60
 bandwidth 64
```

Router R2

```
interface s0/0/0
 ip ospf hello-interval 15
 ip ospf dead-interval 60
```



```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=11ms TTL=126
Reply from 192.168.1.2: bytes=32 time=15ms TTL=126
Reply from 192.168.1.2: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 15ms, Average = 12ms
```

PC2

Desktop Programming

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time=2ms TTL=125
Reply from 172.16.1.2: bytes=32 time=7ms TTL=125
Reply from 172.16.1.2: bytes=32 time=3ms TTL=125
Reply from 172.16.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 3ms
```

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=32ms TTL=126
Reply from 192.168.1.2: bytes=32 time=22ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 32ms, Average = 14ms
```

```
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=22ms TTL=126
Reply from 64.100.1.2: bytes=32 time=14ms TTL=126
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=29ms TTL=126

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 29ms, Average = 16ms
```

PC3

Desktop Programming

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time=15ms TTL=126
Reply from 172.16.1.2: bytes=32 time=6ms TTL=126
Reply from 172.16.1.2: bytes=32 time=10ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 15ms, Average = 8ms
```

```
C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Reply from 172.16.2.2: bytes=32 time=36ms TTL=126
Reply from 172.16.2.2: bytes=32 time=19ms TTL=126
Reply from 172.16.2.2: bytes=32 time=15ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 36ms, Average = 17ms
```

```
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=56ms TTL=125
Reply from 64.100.1.2: bytes=32 time=28ms TTL=125
Reply from 64.100.1.2: bytes=32 time=26ms TTL=125
Reply from 64.100.1.2: bytes=32 time=43ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 26ms, Maximum = 56ms, Average = 38ms
```

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